

Mathematics

Stage 8

Paper 1

2022

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Brackets in mark scheme

When brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6(58...)	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30.....cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e. 18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

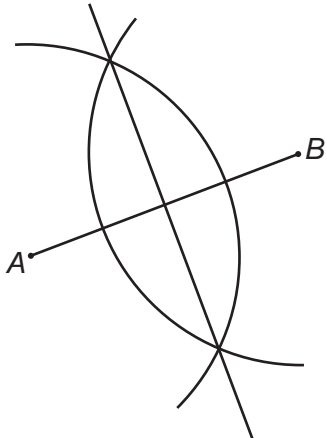
For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

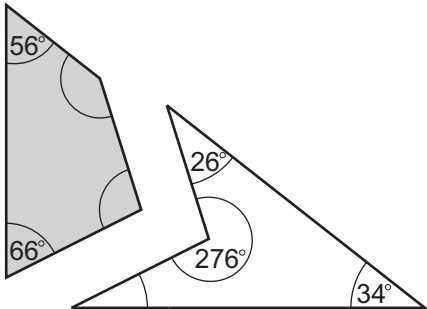
The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance
1	0.7	1		Accept 70% or $\frac{7}{10}$
2	1 10	2	Award 1 mark for each correct answer.	
3	-4	1		
4	(x =) 20	1		
5	20 : 17 correct answer only	1		Do not accept equivalents. Do not accept with units included.
6	$\frac{2}{5}$	2	Award 1 mark for $\frac{12}{15} - \frac{5}{15}$ or for $\frac{7}{15}$ or for answer equivalent to $\frac{2}{5}$ or $\frac{6}{7} \times$ (<i>their</i> subtraction) correctly evaluated and in its simplest form.	
7	(A =) (34, 14) (B =) (27, 0) (C =) (27, 14)	2	Award 1 mark for A or B or C correct.	
8	radius, chord (AB), diameter, circumference	1		All four answers in the correct order for the mark.

Question	Answer	Marks	Part Marks	Guidance
9	-20 100 71	3	Award 1 mark for each correct answer.	
10	$x - 1 > 4$ $5 > x$ $2x > 10$ $x \geq 4$ $5 < x$	1		Accept any clear indication. All three answers correct for the mark.
11(a)	$x = k$, where k is any number.	1		k can be 4
11(b)	A B C D E	1		Accept any clear indication.
11(c)	$y = -x$	1		
12	Accurate bisector with correct arcs. 	2	Award 1 mark for an accurate perpendicular bisector with no or incorrect arcs.	Arcs and accurate bisector must be present for 2 marks. Accuracy: $\pm 2^\circ$ and ± 2 mm for the midpoint.

Question	Answer	Marks	Part Marks	Guidance						
13	60 (minutes) 11 42 ± 3 minutes 135 (km) ± 3 km	2	Award 1 mark for two correct answers.							
14(a)	<table border="1"><tr><td>$n + k$ or $n - k$</td><td>$n \times k$</td></tr><tr><td>(A)</td><td>B</td></tr><tr><td>D</td><td>C</td></tr></table>	$n + k$ or $n - k$	$n \times k$	(A)	B	D	C	1		All three answers correct for the mark.
$n + k$ or $n - k$	$n \times k$									
(A)	B									
D	C									
14(b)	799	1								
15	$6\frac{1}{3}$ correct answer only	2	Award 1 mark for $\frac{76}{12}$ or $\frac{19}{3}$ or equivalent.	Or equivalent, e.g. $6\frac{4}{12}$ or $6.\dot{3}$ for 1 mark.						

Question	Answer	Marks	Part Marks	Guidance
16(a)	Ask every member who comes to the gym on Tuesday morning. ☐ Use a random number generator to generate 50 membership numbers and ask members with those numbers. ☒ Ask every 10th member who comes to the gym during one week. ☐ Call all members who haven't been to the gym for a month to ask them. ☐	1		Accept any clear indication.
16(b)	Accept any correct explanation, e.g. - The sample size is small. - The sample may not represent the population (because of sampling methods).	1		
17		3	Award 2 marks for four correct angles or award 1 mark for two correct angles.	Ignore degrees symbol.
18	241, 243, 245 421, 423, 425	2	Award 1 mark for one missing or one repeated.	Must be no repeats and none missing for 2 marks.

Question	Answer	Marks	Part Marks	Guidance												
19(a)	−26.95	1														
19(b)	0.9	1														
20	(<i>x</i>) decimal between 0.45 and 0.5 and (<i>y</i>) fraction between $\frac{12}{24}$ and $\frac{14}{24}$ e.g. $\frac{13}{24}$	2	Award 1 mark for (<i>x</i>) decimal between 0.45 and 0.5 or (<i>y</i>) fraction between $\frac{12}{24}$ and $\frac{14}{24}$ e.g. $\frac{13}{24}$													
21	(−4, 1)	1														
22	<table><tr><th>Number of vertices</th><th>Number of faces</th><th>Number of edges</th></tr><tr><td>(4)</td><td>(4)</td><td>(6)</td></tr><tr><td>(12)</td><td>20</td><td>(30)</td></tr><tr><td>(<i>v</i>)</td><td>(<i>f</i>)</td><td><i>v</i> + <i>f</i> − 2</td></tr></table>	Number of vertices	Number of faces	Number of edges	(4)	(4)	(6)	(12)	20	(30)	(<i>v</i>)	(<i>f</i>)	<i>v</i> + <i>f</i> − 2	2	Award 1 mark for each correct answer.	
Number of vertices	Number of faces	Number of edges														
(4)	(4)	(6)														
(12)	20	(30)														
(<i>v</i>)	(<i>f</i>)	<i>v</i> + <i>f</i> − 2														
23	1 014 049 1009 1 024 144	2	Award 1 mark for two correct answers.													

Question	Answer	Marks	Part Marks	Guidance
24	$(x =) 5$	2	Award 1 mark for $36 = 2^2 \times 3^2$ or $56 = 2^3 \times 7$ or for $\frac{36 \times 56}{3^2 \times 7}$ or equivalent or better or for answer 2^5	e.g. $\frac{2016}{63}$ or $\frac{4 \times 8}{1 \times 1}$ or 32
25	Any correct criticism, e.g. - The shapes representing the percentages are not in proportion to percentages. - The image is in 3D meaning it looks like volumes represent percentages.	1		Accept, e.g. science fiction is a bigger shape than action but represents a smaller percentage.
26	-135	2	Award 1 mark for 45×-3 or equivalent or for 45×-2 or equivalent or for answer 135	Implied by -90
27	h (is greater by) 1	1		
28	$6y$	2	Award 1 mark for $3y(2y - 5)$ or area of rectangle is $12y^2 - 30y$ or equivalent.	